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CASE STUDY — 1

BrightMart is a medium sized online shopping company. Custombers use its website and mobile application to purchase product

The company stores.

1. Customber Names
2. email addresses
3. Phone numbers
4. Delivery addresses
5. Passwords
6. Order information.
7. Payment - related informations.

Brightmart has around 100 employees.

The company recently hired several new employees. To make things easier, the IT department gave employees access to different internal systems.

However, the company has the following security weaknesses.

1. Some employees use simple passwords such as BrightMart123.
2. Many employees use the same passwords for multiple systems.
3. Software updates are sometimes developed.
4. The company does not regularly conduct employee

cybersecurity training.

5. The company does not regularly conduct employee cybersecurity training.
6. There is no multi-factor authentication for some important accounts.
7. Employees can access sensitive information that they do not necessarily need for their jobs.
8. Backups are performed only once a month.

8.

The incident

Once Monday morning, an employee named Rahul receives an email.

Subject: Urgent - Your Account will be Suspended.

The email appears to come from the company's IT department.

It says:

"Your company account will be suspended today. Click the link below and verify your password immediately."

Rahul clicks the link.

The page looks almost identical to the company's login page.

He enters his username and password.
Rahul doesnot realize that the website is controlled by an attacker.
The attacker now has Rahul's login credentials.

Stage 1: — Unauthorized Access

The attacker uses Rahul's creditienals to access Brightmart's internal system.
Because Rahul uses the same passwords for another system, the attacker is also able to access that system.
The attacker discovers that Rahul has access to customber information.
The attacker downloads a large amount of ceustomber data.

✧ (What cybersecurity problem has occurred?

Stage 2 — Malware attack.

The attacker sends another email to Rahul containing a malicious attachment.

Rahul opens the attachment.
Malware is installed.

The malware begins communication with attackers system.
The attacker now attempts to move from Rahul's computer to another systems within Brightmart's network.

Question —

Which Vulnerability allowed the attacker to move further into the organization?

Stage — 3 Ransomware.

Two days later, employees arrive at work and discover that they cannot access important files.

A message appears on their computer.

"Your files have been encrypted. Pay the ransom to recover them."

The company's customer database and several important business files are encrypted.
Unfortunately, the company's most recent backup is almost a month old.
BrightMart's website stops working.
Customers cannot place orders.
The customer-support team cannot access customer records.

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Stage 4 – Business IT Impact

Bright Mart now faces several problems.

Financial Impact
The company loses sales because its website is unavailable.

Data Impact
Customber information may have been stolen.

Operational Impact.
Employees cannot access important systems.

Reputation Impact
Custombers lose confidence in the company.

Legal / Compliance Impact
The company may face regulatory or legal consequences – depending on what information was exposed and which laws apply.

Group – 1
Identify the Threats.
What threats are present in this case

The threats present in the Bright Mart case are:

1 Phishing attack — The attacker sent a fake email to Rahul and stole his login credentials.

2 Malware attack — A malicious attachment installed on Rahuls computer.

3. Ransomware attack — The attacker encrypted BrightMarts files and customber database and demanded for a ransom.

4. Unauthorized Access — The attacker used Rahuls stolen credentials to access BrightMarts internal systems.

5 Credential theft

6. Group - 2 — Identify the Vulnerabilities.

What weaknesses in BrightMarts security allowed the attack to succeed?

The main weaknesses in BrightMarts security were:

1 Weak passcoords — Employees used simple passwords such as BrightMart123.

2 Passcoord reuse - Rahul used the same password for multiple systems, allowing the attacker to get the access.

3 Lack of cybersecurity training - Employees were not regularly trained to recognize phishing mails.

4. Delayed ~~pas~~ Software updates - Outdated software may contain security vulnerabilities that attackers can exploit.

5. Excessive privileges — Attacker can access more data.

Group 3 – Apply the CIA Triad.

Was confidential information exposed? (Confidentiality)
yes, the confidential information was exposed. The attacker stole Rahul's credentials and downloaded customber information.

Was information changed or damaged? (Integrity)
yes, the informated was changed, damaged. Ransomware encrypted important files and the customber database, making the original information damaged. ~~(Confidentiality)~~.

Were systems available to legitimate users? ~~(Integrity)~~
The website and internal systems became unavailable, so custombers could not place orders and employees could not access customber records. (Availability).

Group 4. Security Solutions.

Q1. Who is responsible for cybersecurity.
— Everyone in the organization is responsible for cyber-security.
Employees should use strong passwords, recognize phishing and report suspicious activity.
System administrators should ensure employees only have the access they need.

Q2. Was phishing the only problem?
— Nay phishing was the attackers initial methods, but several security weaknesses allowed the attack to

(the attack could have been prevented) become more serious.

Q3. What was the biggest vulnerability?

The biggest vulnerability was the combination of weak / re-used passwords, because stolen credentials could then be used to access multiple systems.

Lack of employee awareness was a major vulnerability beause the attack started when Rahul clicked the phishing link.

Jawa 3/4/26

Attackers exploit systems, network and human psychology.

Dos and Dos (Denial - of - service).

Overwhelming a target server with massive traffic from compromised systems until legitimate users cannot access the website.

Man in the middle. (MITM).

Positioning secretly between communicating parties to intercept or manipulate data.

Social engineering and Phishing.

Bypassing technical defenses entirely by exploiting human trust, fear or urgency to steal passwords.

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Phases of HACKING

1. Footprinting - Collection of information of victim as much as possible.
2. Scanning - Checking wheather information collected is right or wrong.
3. Graining access - To victims compater or system
4. Maintaining the access - maintaining the access till your goal is achieved.
5. Cleaning tracks - Cleaning the evidence of your existence.

✧ The Anatomy of Attacks (cyber)

✱ Types of Attacks:

1. White Hat Attackers. → Attacks to secure (Legal)
2. Black Hat Attacker → Find Vulnerability then report to concerned authority and ask money. If they fail to give money they leak the data. Second job of Black Hat Hackers is to send numerous request at a time to the server inorder to put the server or website down